METRIC

DOD-G-51510
26 July 1985

MILITARY SPECIFICATION

GOGGLES, INDUSTRIAL, EYECUP AND LENSES, GOGGLES, INDUSTRIAL (METRIC)

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types and two classes of eyecup industrial goggles and three types of lenses for the goggles.

1.2 Classification. Goggles and lenses shall be of the following types and classes as specified (see 6.2).

1.2.1 Goggles.

Type I - Cup-type
  Class A - Chipper's
  Class B - Welder's and cutter's

Type II - Cover-cup-type
  Class A - Chipper's
  Class B - Welder's and cutter's

1.2.2 Lenses.

Type I - Clear
Type II - Filter
Type III - Cover

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Armament, Munitions and Chemical Command, ATTN: AMSMC-TDC-S (A), Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4240

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

UU-P-553	-	Paper, Wrapping, Tissue
PPP-B-566	-	Boxes, Folding, Paperbound
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-591	-	Boxes, Shipping, Fiberboard, Wood-Cleated
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-665	-	Boxes: Paperboard, Metal Edged and Components
PPP-B-676	-	Boxes, Setup
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing
PPP-T-76	-	Tape, Packaging, Paper (For Carton Sealing)

MILITARY

MIL-L-10547	-	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are indicated as DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) STANDARDS

ANSI Z87.1 - Practice for Occupational and Educational Eye and Face Protection (DOD Adopted)

(Application for copies should be addressed to The American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

ASTM STANDARDS

D 3951 - Commercial Packaging (DOD Adopted)

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. When tested as specified in 4.2.4.1, materials used in the manufacture of goggles and lenses shall conform to the ANSI Z87.1 requirements under the section on Protectors - Materials and Methods of Test for materials, disinfection, corrosion-resistance, and flammability.

3.2 Goggle components. Goggle components shall conform to the ANSI Z87.1 general requirements for eyecup goggles.

3.2.1 Nose bridge. The nose bridge shall be flexible and shall be easily adjustable by hand to a fixed position. Type II, class B goggles shall be equipped with a leather nose bridge pad of sufficient size, between the eyecups, to protect the eyes in the vicinity of the nose from harmful radiation.

3.2.2 Headband. The headband shall be made of solid, oil-resistant synthetic rubber no less than 9.5 millimeters (mm) wide. It shall be adjustable in length, shall be fastened to the eyecups in such a manner as to be easily replaceable, and shall not interfere with the facial fit of the eyecups.

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3.2.3 Eyecups.

3.2.3.1 Materials. When tested as specified in 4.2.4.1, eyecups shall conform to the ANSI Z87.1 detailed requirements for eyecup material.

3.2.3.2 Vision and fit. Eyecups shall conform to the ANSI Z87.1 detailed requirements for vision and fit for cup-type goggles (type I) or cover-cup-type goggles (type II), as applicable.

3.2.3.3 Ventilation. Eyecups shall conform to the ANSI Z87.1 detailed requirements for ventilation for chipper's models (class A) or welder's and cutter's models (class B), as applicable.

3.2.3.4 Lens-retaining rings. Eyecups shall conform to the ANSI Z87.1 detailed requirements for lens-retaining rings.

3.2.3.5 Lens seat. Eyecups shall conform to the ANSI Z87.1 detailed requirements for lens seat.

3.2.3.6 Interior finish. The eyecups shall have a permanent, dull, low-reflecting interior finish.

3.2.4 Lenses. Each pair of class A goggles shall be furnished with two type I lenses. Each pair of class B goggles shall be furnished with two type II filter lenses of the shade specified (see 6.2) and two type III lenses. Lenses shall conform to the following ANSI Z87.1 requirements, when tested as specified in 4.2.4.1 or examined as specified in 4.2.3.1, as applicable.

3.2.4.1 ANSI Z87.1 detailed requirements for eyecup goggle lenses.

(a) Dimensions
(b) Optical quality
(c) Impact resistance
(d) Plastic lens penetration (if applicable)
(e) Edges
(f) Transmittance
(g) Haze
(h) Cover lenses
(i) Marking

3.2.4.2 ANSI Z87.1 general requirements for lenses.

(a) Optical quality
(b) Prismatic power
(c) Refractive power
(d) Astigmatism
(e) Definition
(f) Size tolerance
(g) Edges
(h) Haze

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3.2.4.3 ANSI Z87.1 detailed requirements for lenses.

(a) Lens thickness
(b) Marking
(c) Transmittance
(d) Lens strength

3.3 Shades of filter lenses. Filter lenses shall be of the shades in the ANSI Z87.1 table entitled Transmittance and Tolerances in Transmittance of Various Shades of Absorptive Lenses, Filter Lenses, and Plates (see 6.2).

3.4 Sodium light filter lenses. When required, sodium light filter lenses shall be furnished in the shades shown in table I as specified (see 6.2). Sodium light filter lenses shall have a strong absorption band in the region of the sodium line (589.3 nanometers). The lenses shall meet the requirements for type II filter lenses except that the thickness of the lenses shall be no less than 2.2 mm and no more than 4.0 mm. When tested as specified in 4.2.4.2, the transmission at 589.3 nanometers shall be no greater than 15 percent of the total visible transmission, as shown in table I. The maximum transmission shall be between 485 and 600 nanometers.

TABLE I. Sodium filter lens light transmittance

Shade number	Percent sodium light (589.3 nanometers) transmittance		
	Minimum	Standard	Maximum
2.5	2.70	3.45	4.34
3.0	1.31	2.09	2.69
4.0	0.49	0.78	1.28
5.0	0.18	0.29	0.47
6.0	0.07	0.11	0.18

3.6 Workmanship. Eyecup goggles, components, and lenses shall be free from defects such as tears, cracks, and chips, and shall be free from foreign matter such as dirt, oil, and grease.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the goggles of one type and class or the replacement lenses of one type produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions, and offered for delivery at one time. However, no more than one lot of each type of lens shall be represented in any one lot of goggles.

4.2.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.3 Inspection procedure.

4.2.3.1 For examination of goggles and lenses. The sample unit shall be one pair of goggles with applicable lenses or one pair of replacement lenses, as applicable. Sample goggles and lenses shall be examined for the following applicable defects using an AQL of 1.0 percent defective for each individual defect.

(a) Goggles not as specified
(b) Goggle components not assembled as specified or missing
(c) Nosebridge not as specified
(d) Headband not as specified
(e) Vision and fit of eyecups not as specified
(f) Ventilation of eyecups not as specified
(g) Lens-retaining ring not as specified
(h) Lens seat not as specified
(i) Interior finish of eyecups not as specified
(j) Lenses not as specified
(k) Dimensions of lenses incorrect
(l) Lenses not free of visible surface defects
(m) Edges of lenses not as specified
(n) Marking of lenses incorrect, illegible, or missing
(o) Filter lenses not of the shade specified
(p) Workmanship not as specified

4.2.3.2 For examination of packaging. The sample unit shall be one filled unit, intermediate, or shipping container, as applicable, ready for shipment. Sample unit, intermediate, and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

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(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or broken
(e) Tissue paper missing or not as specified
(f) Shipping container caseliners missing or incorrect
(g) Marking incorrect, missing, or illegible

4.2.3.3 For test of goggles and lenses. The sample unit shall be one pair of goggles with applicable lenses or one pair of replacement lenses, as applicable. Sample goggles and lenses shall be subjected to all applicable tests specified in 4.2.4 using an AQL of 0.65 percent defective. Any pair of goggles (with lenses) or any pair of replacement lenses that fails to pass all the required tests shall be considered a defective item.

4.2.4 Tests. Tests shall be conducted as follows:

4.2.4.1 ANSI Z87.1 tests. Conduct the following tests in accordance with the applicable ANSI Z87.1 procedures:

(a) Ability of materials to withstand disinfection
(b) Corrosion-resistance of any metal parts
(c) Flammability of plastic materials
(d) Heat deformation of eyecups
(e) Prismatic power of lenses
(f) Refractive power and astigmatism of lenses
(g) Definition of lenses
(h) Haze of lenses
(i) Ultraviolet, luminous (visible), and infrared transmittance of lenses
(j) Impact resistance of lenses
(k) Penetration test for plastic lenses

4.2.4.2 Sodium light transmittance (for sodium light filter lenses only). Determine the transmittance of the lenses at the sodium wavelength by means of a suitable spectroradiometer and a sodium arc lamp or other lamp producing monochromatic light at 589.3 nanometers.

5. PACKAGING

5.1 Preservation (level A, B, or commercial). Goggles and lenses shall be preserved level A, B, or commercial as specified (see 6.2).

5.1.1 Levels A and B.

5.1.1.1 Unit packing.

(a) Goggles. Goggles shall be individually unit packed in paperboard folding, set-up, or metal-edge boxes conforming to PPP-B-566, PPP-B-676, or PPP-B-665, respectively. Lenses shall be protected from scratches by

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wrapping them with tissue paper conforming to type II of UU-P-553. Sufficient cushioning shall be provided to prevent movement and damage to the goggles within the box during handling and shipment.

(b) Replacement lenses. Lenses of the same type, when purchased separately, shall be individually wrapped in tissue paper conforming to type II of UU-P-553, and shall be cushioned in pairs. Each pair shall be unit packed and sealed in a kraft paper envelope with an adhesive covered flap.

5.1.1.2 Intermediate packing.

(a) Goggles. Goggles unit packed as specified in 5.1.1.1(a) shall be intermediately packed in quantities of 10 in folding, set-up, or metal-edge boxes conforming to PPP-B-566, PPP-B-676, or PPP-B-665, respectively. Boxes shall be closed and sealed in accordance with the applicable box specification or appendix thereto.

(b) Replacement lenses. Lenses unit packed in pairs as specified in 5.1.1.1(b) shall be intermediately packed in quantities of 12 pairs in boxes as specified in 5.1.1.2(a).

5.1.2 Commercial. Goggles and lenses shall be unit packed in accordance with ASTM D 3951.

5.2 Packing (level A, B, or commercial). Goggles and lenses shall be packed level A, B, or commercial as specified (see 6.2).

5.2.1 Level A. Goggles, or lenses when acquired separately, shall be packed in a container conforming to class 3 of PPP-B-585, type overseas of PPP-B-591, type overseas of PPP-B-601, class 2 of PPP-B-621, or class weather-resistant of PPP-B-636. Shipping containers shall be provided with caseliners conforming to MIL-L-10547. Caseliners shall be closed and sealed in accordance with the appendix to MIL-L-10547. Caseliners for fiberboard boxes conforming to PPP-B-636 may be omitted provided all corners and edge seams and manufacturer's joints are sealed with minimum 38 mm wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto, except fiberboard boxes may be banded with filament reinforced tape in accordance with the appendix to PPP-B-636. The gross weight of wood or cleated-wood boxes shall not exceed 91 kilograms.

5.2.2 Level B.

5.2.2.1 Goggles (other than plastic). Ten boxes intermediately packed as specified in 5.1.1.2(a), containing goggles of the same type and class, shall be packed in a fiberboard box conforming to type CF or SF, class domestic, grade equal to a bursting strength of not less than 275 of PPP-B-636. Closure shall be by adhesive applied over at least 75 percent of the flap area or by the two strip tape method specified in Rule 41, Section 7 of the Uniform Freight Classification Rules, using tape conforming to PPP-T-45.

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5.2.2.2 Goggles (plastic). Seventy-two individually boxed plastic goggles shall be packed in a container as specified in 5.2.2.1.

5.2.2.3 Replacement lenses. A specified quantity of lenses of the same type (see 6.2), intermediately packed as specified in 5.1.1.2(b), shall be packed in a container as specified in 5.2.2.1.

5.2.3 Commercial. Goggles and lenses shall be packed in accordance with ASTM D 3951.

5.3 Marking. Unit, intermediate, and shipping containers shall be marked in accordance with MIL-STD-129 for level A and B shipments and in accordance with ASTM D 3951 for commercial shipments.

6. NOTES

6.1 Intended use. Type I goggles are intended for use by persons who do not wear eyeglasses. Type II goggles are intended for use over eyeglasses. Class A goggles are intended to protect the wearer's eyes from flying objects. Class B goggles are intended to protect the wearer's eyes from glare, injurious radiations, and impact. Cover lenses are intended as a protection against pitting of the filter lenses. Sodium light filter lenses are intended for welding or cutting operations where the fluxes are rich in sodium compounds, such as in aluminum welding. Shade numbers 1.7 to 3.0 filter lenses are intended for use in controlling reflected sunlight glare from snow, water, sand, and other surfaces. Suggested shade numbers for various welding and cutting operations are contained in ANSI Z87.1 under the table entitled Selection of Shade Number for Welding Filters.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Type and class of goggles required (see 1.2.1)
(c) Shade number required for filter lenses (see 3.2.4 and 3.3)
(d) Shade of sodium light filter lenses, when required (see 3.4)
(e) Level of preservation and packing required (see 5.1 and 5.2)
(f) Quantity of replacement lenses required in level B packing box (see 5.2.2.3)

6.3 Specification replacement data. This specification is a replacement for GGG-G-513B, dated September 11, 1970, and Amendment 1, dated May 28, 1975. These documents were canceled March 2, 1984.

6.4 Cross-reference to ANSI Z87.1 requirements. A cross-reference between the requirements of this specification which reference ANSI Z87.1 and the requirement paragraphs of ANSI Z87.1-1979 is provided in table II. The ANSI Z87.1 paragraph numbers may change when the 1979 version of the standard is revised.

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TABLE II. Cross-reference for ANSI Z87.1 requirements

DOD-G-51510 requirement paragraph	ANSI Z87.1-1979 requirement paragraph
3.1	6.2
3.2	6.1.1.3.1
3.2.3.1	6.1.1.4.1
3.2.3.2	6.1.1.4.3
3.2.3.3	6.1.1.4.4
3.2.3.4	6.1.1.4.5
3.2.3.5	6.1.1.4.6
3.2.4.1	6.1.1.4.7
3.2.4.2	6.3.2
3.2.4.3	6.3.3

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE
DOD-G-51510	GOGGLES, INDUSTRIAL, EYECUP AND LENSES, GOGGLES, INDUSTRIAL (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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